Introduction to Artificial Intelligence

Topic 8: Machine Learning in Python

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Wat is Machine Learning?

En wat is het verschil met AI?

En met traditioneel programmeren?

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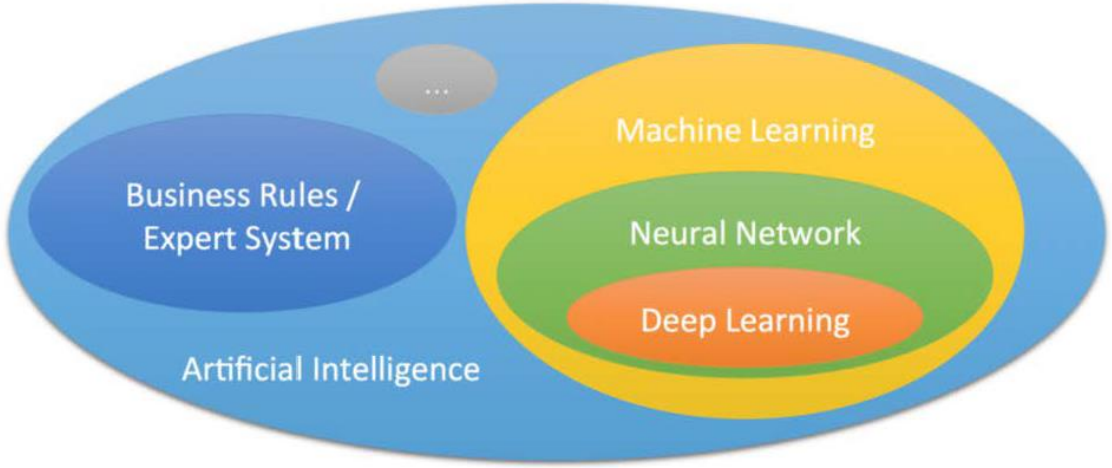
AI – ML - DL

- **Artificial Intelligence** (AI):

  "The set of all tasks in which a computer can make decisions."
- **Machine Learning** (ML):

  "The set of all tasks in which a computer can make decisions *based on data*."
- **Deep Learning** (DL):

  "The field of machine learning that uses certain objects called *neural networks*."



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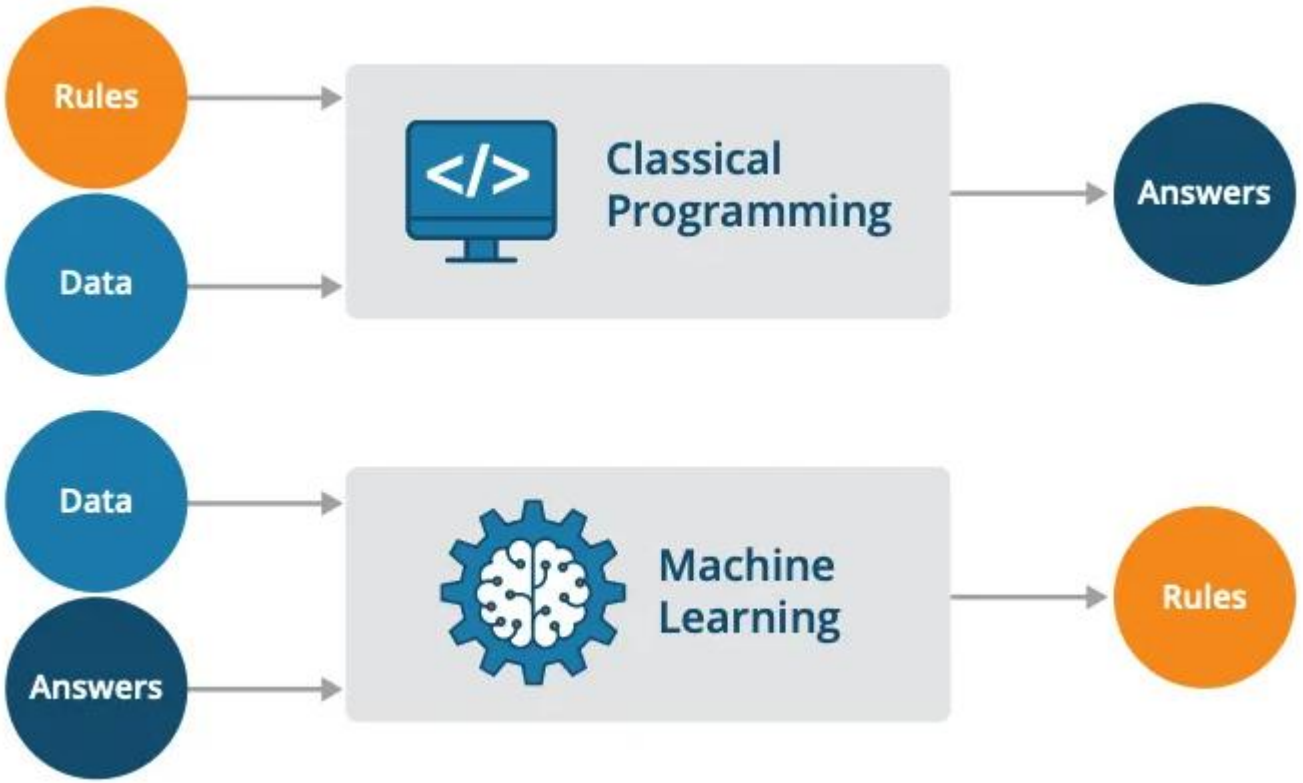
Machine Learning

Voorbeeld: nieuwe auto kopen

- Hoe nemen wij beslissingen?
  - Door logisch **redeneren**
  - Door gebruik te maken van **ervaringen** (van onszelf of van anderen)
- Voor een computer: **ervaringen = data**

"Machine learning is common sense, except done by a computer."

ML vs traditioneel programmeren



Model vs Algoritme

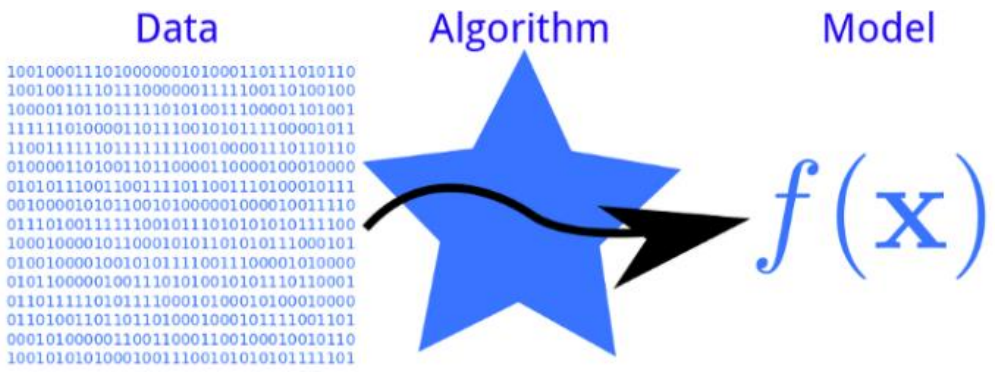
- **Model**:

  "A set of rules that represent our data and can be used to make predictions."

- **Algoritme**:

  "A procedure, or a set of steps, used to build a model."

"An algorithm is run on data to create a model"



Verschillende types Machine Learning

Wat zijn de drie grote families?

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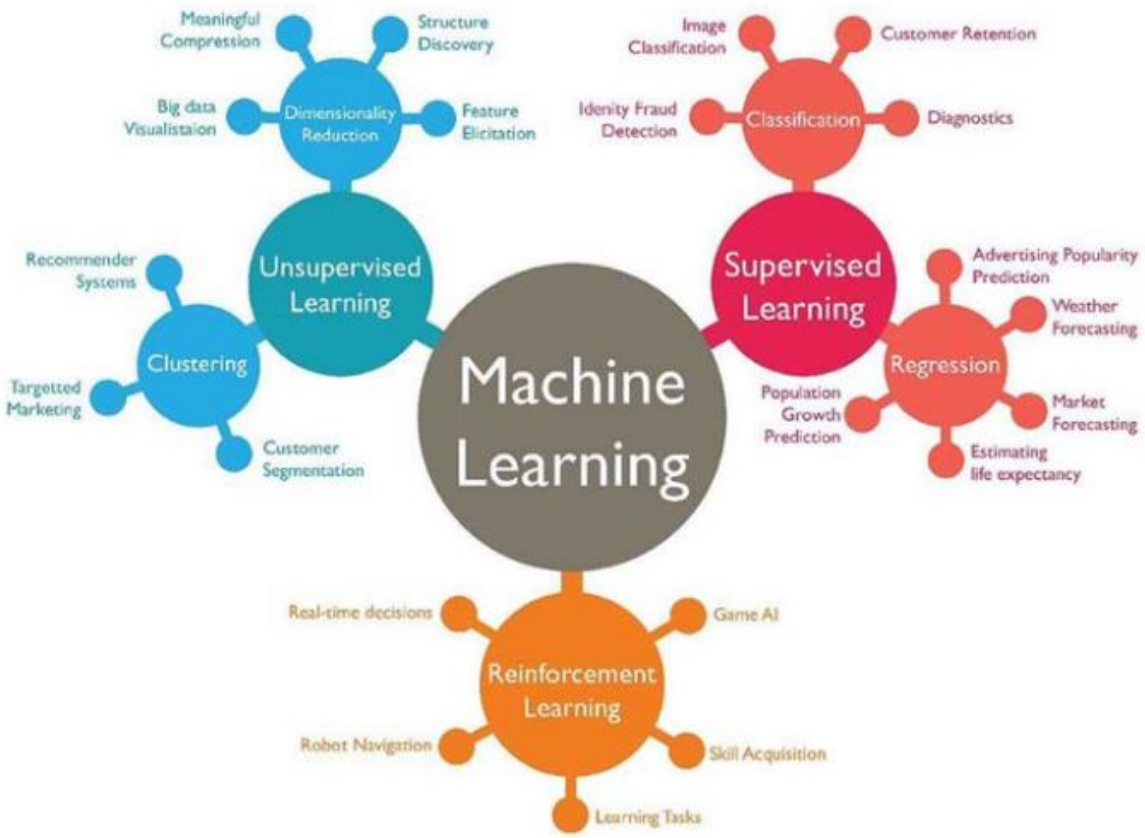
De 3 grote families



Category of machine learning. Image by https://www.techleer.com/articles/203-machine-learning-algorithm-backbone-of-emerging-technologies/

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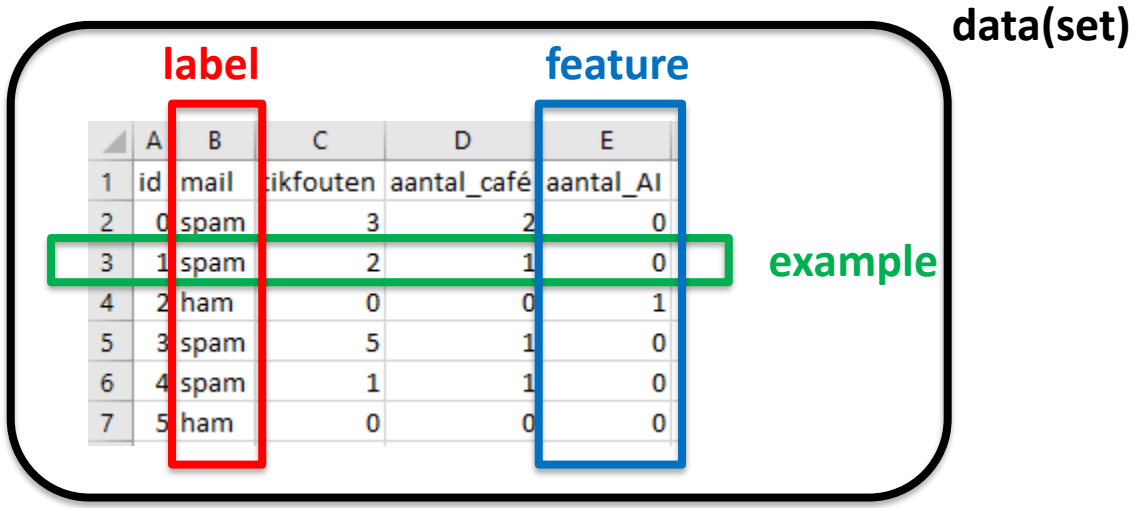
Data, examples, features en labels

- **Data** = informatie, bijv. tabel
- **Example** = datapunt = rij in de tabel
- **Feature** = kolom in de tabel
- **Label** = feature die je wil voorspellen = target

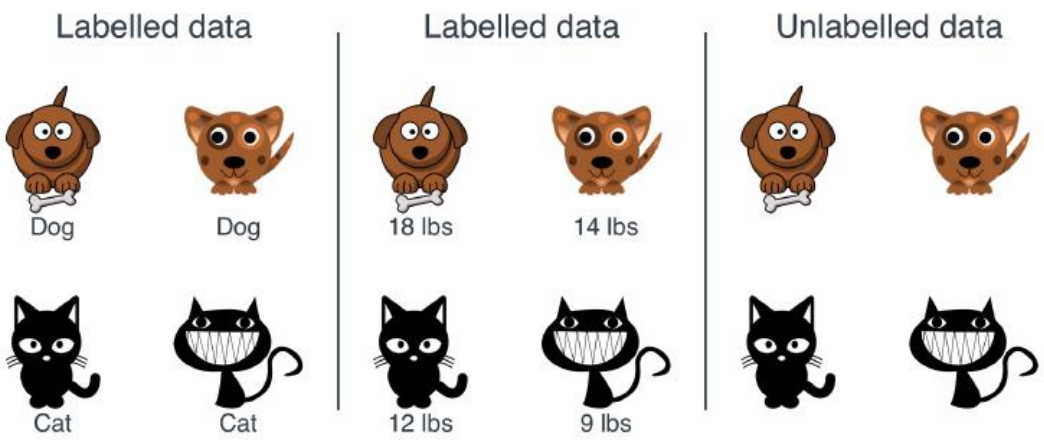


	A	B	C	D	E
1	id	mail	tikfouten	aantal_café	aantal_AI
2	0	spam	3	2	0
3	1	spam	2	1	0
4	2	ham	0	0	1
5	3	spam	5	1	0
6	4	spam	1	1	0
7	5	ham	0	0	0

Gelabelde en niet-gelabelde data

- **Gelabelde data**: data met een label
  → **SUPERVISED** LEARNING

- **Niet-gelabelde data**: data zonder label
  → **UNSUPERVISED** LEARNING

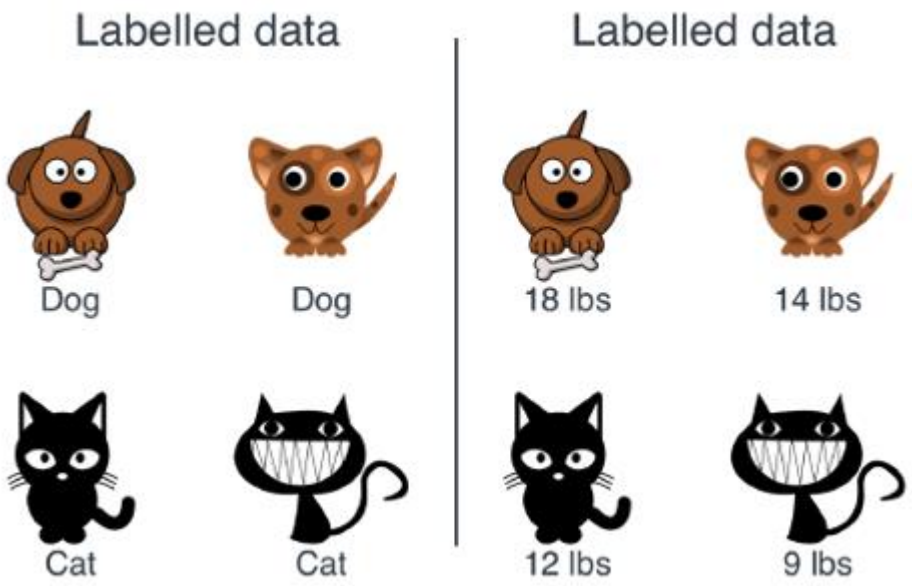


Supervised Learning

Werken met gelabelde data

Numerieke en categorische data

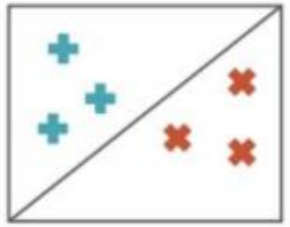
- **Categorische (= kwalitatieve) data**
- **Numerieke (= kwantitatieve) data**



Categorisch **Numeriek**

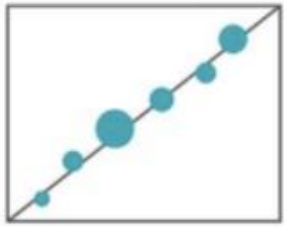
Classificatie en regressie

- **Classificatie**:
  - Modellen die **categorische** data voorspellen
  - Modeloutput is **discreet**: een categorie, een klasse, een toestand, …
  - Voorbeelden: spam of ham, images met kat of hond, MNIST cijfers
- **Regressie**:
  - Modellen die **numerieke** data voorspellen
  - Modeloutput is **continu**: een reëel getal
  - Voorbeelden: prijzen van huizen voorspellen, temperatuur voorspellen



CLASSIFICATION

Sorting items into categories



REGRESSION

Identifying real values (dollars, weight, etc.)

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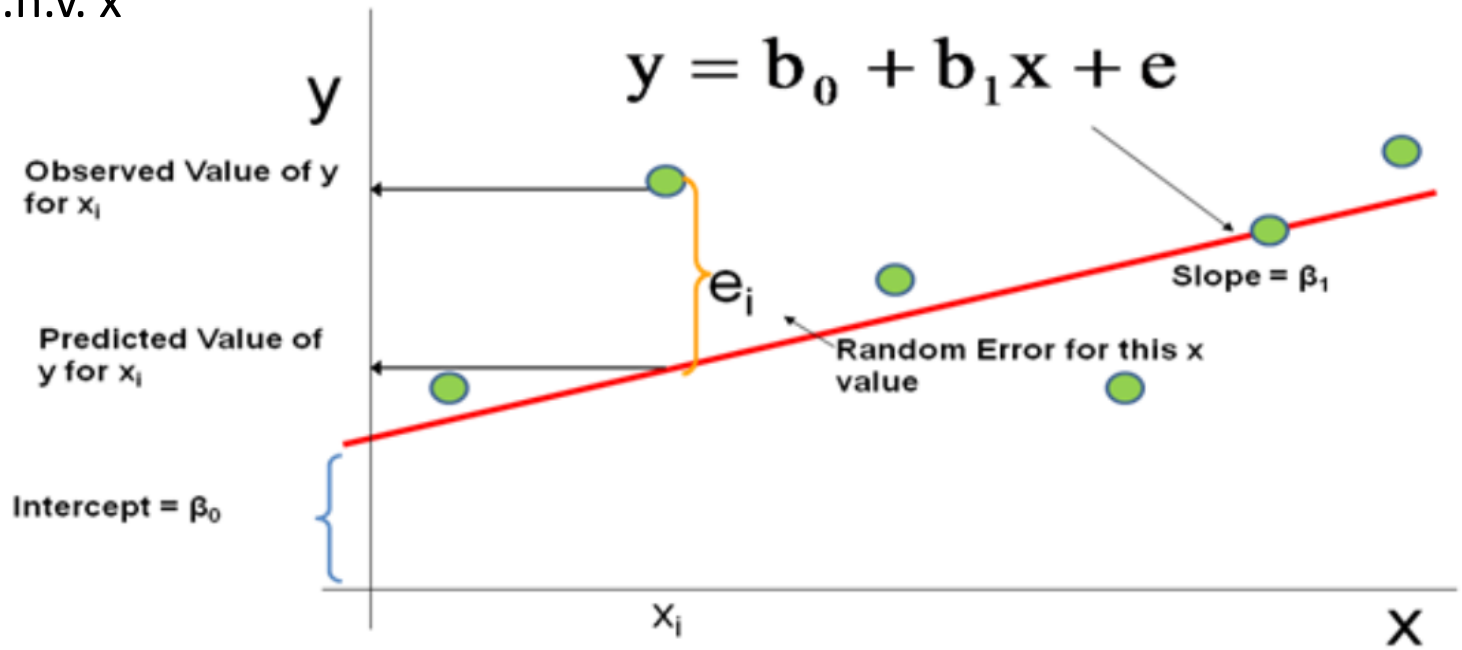
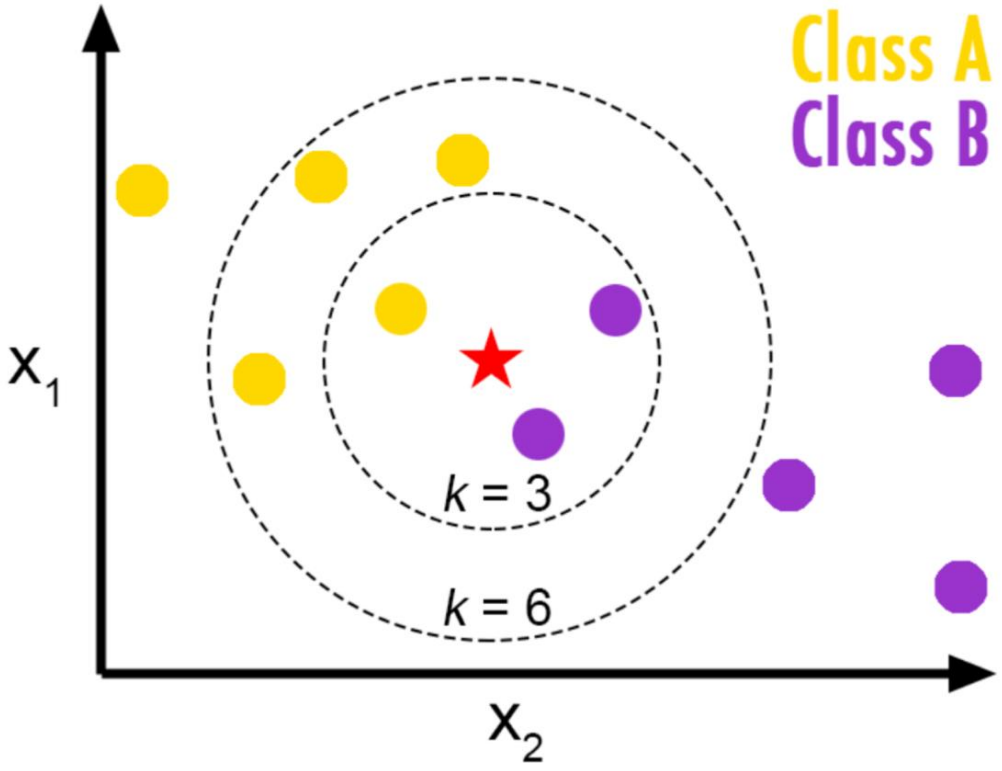
Voorbeeld: enkelvoudige lineaire regressie

- 1 feature x en 1 target y
- lineair verband tussen x en y
- best passende rechte zoeken
- totale error minimaliseren
- y voorspellen a.d.h.v. x



Voorbeeld: K Nearest Neighbors

- features x_1 en x_2
- 2 klassen: geel of paars label
- nieuw punt (ster): welke klasse?
- k dichtste punten bekijken

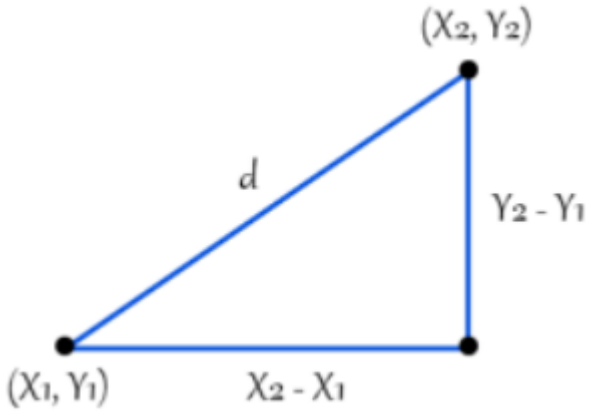


Voorbeeld: K Nearest Neighbors

Hoe de K dichtste punten bepalen?

→ Afstanden berekenen!

- Manhattan afstand *m*
- Euclidische afstand *d*



$$m = |x_2 - x_1| + |y_2 - y_1|$$

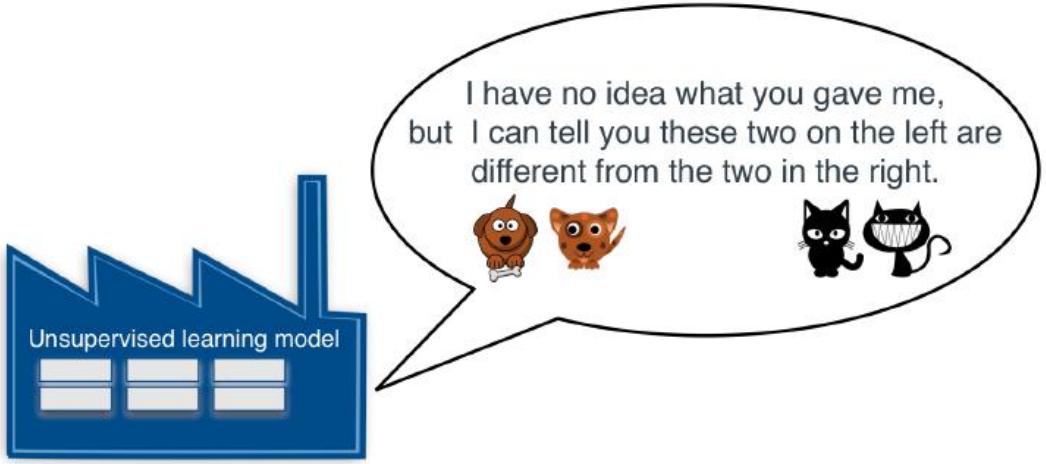
$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Unsupervised Learning

Werken met niet-gelabelde data

Clustering en Dimensionality Reduction

- **Clustering**: data groeperen o.b.v. gelijkenissen



- **Dimensionality Reduction**: data beschrijven met minder features zonder veel verlies aan informatie

Beiden vaak toegepast tijdens **preprocessing** van data

Clustering

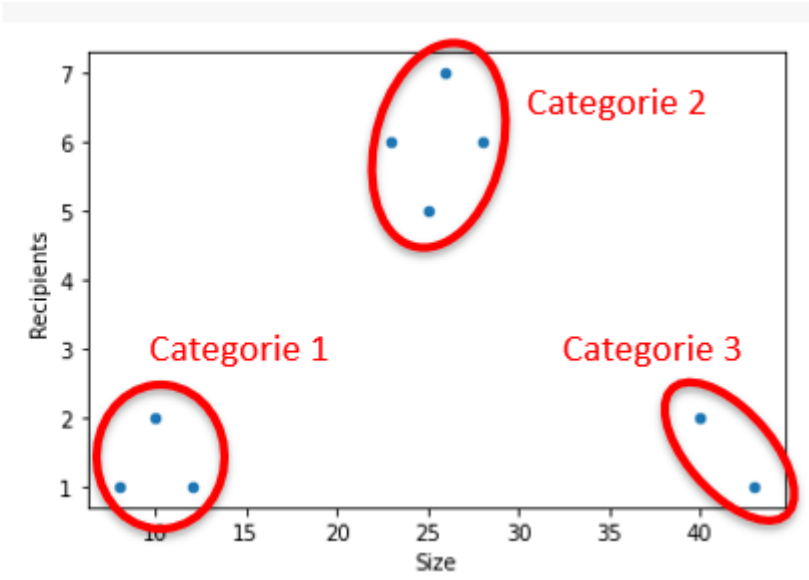
Voorbeelden:

- Genetica: soorten groeperen obv gelijkenissen
- Medische beeldvorming: beelden opdelen obv weefselstructuren
- Markt segmentatie: klanten groeperen obv demografie, inkomen, …
- Mails:

Geen labels!

E-mail	Size	Recipients
1	8	1
2	12	1
3	43	1
4	10	2
5	40	2
6	25	5
7	23	6
8	28	6
9	26	7

mails.csv



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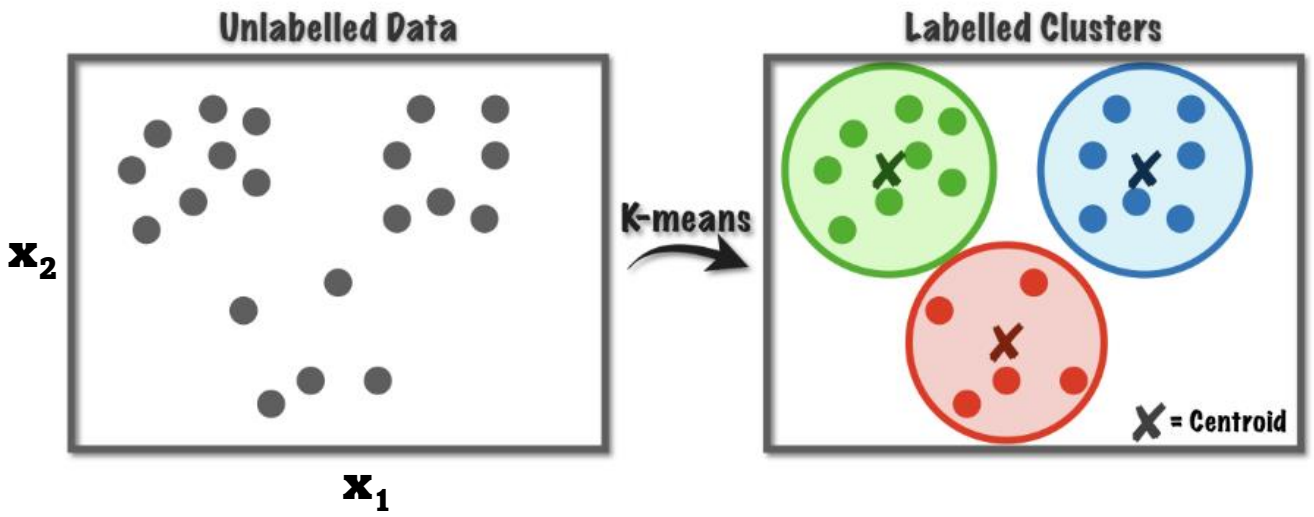
Voorbeeld: K-Means Clustering

- features x_1 en x_2
- geen klassen!
- iteratief datapunten opdelen in k clusters adhv k centroïden
- https://youtu.be/nXY6PxAaOk0



Deep Learning

Werken met artificiële neurale netwerken

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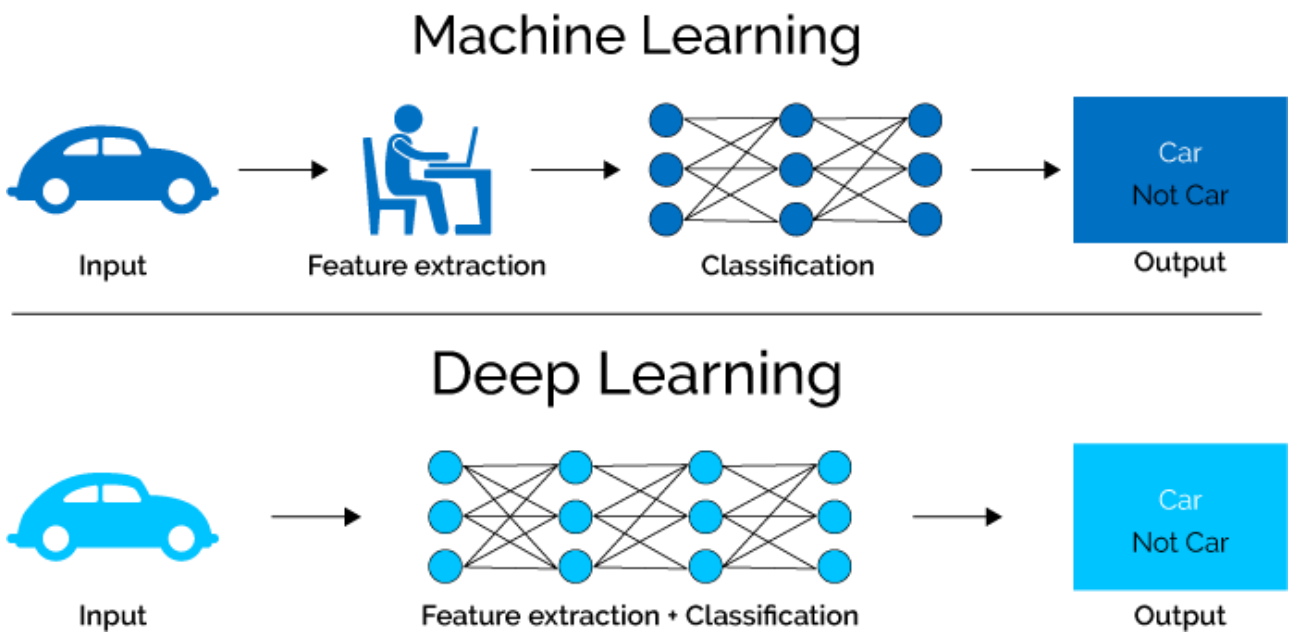
Wat is deep learning?

- Subfield of ML for **learning representations** of data.
- Exceptional effective at **learning patterns**.
- Utilizes learning algorithms that derive **meaning** out of data by using a **hierarchy of multiple layers** that mimic the **neural networks of our brain**.
- If you provide the system **tons of information**, it begins to understand it and respond in useful ways.



https://www.xenonstack.com/blog/static/public/uploads/media/machine-learning-vs-deep-learning.png

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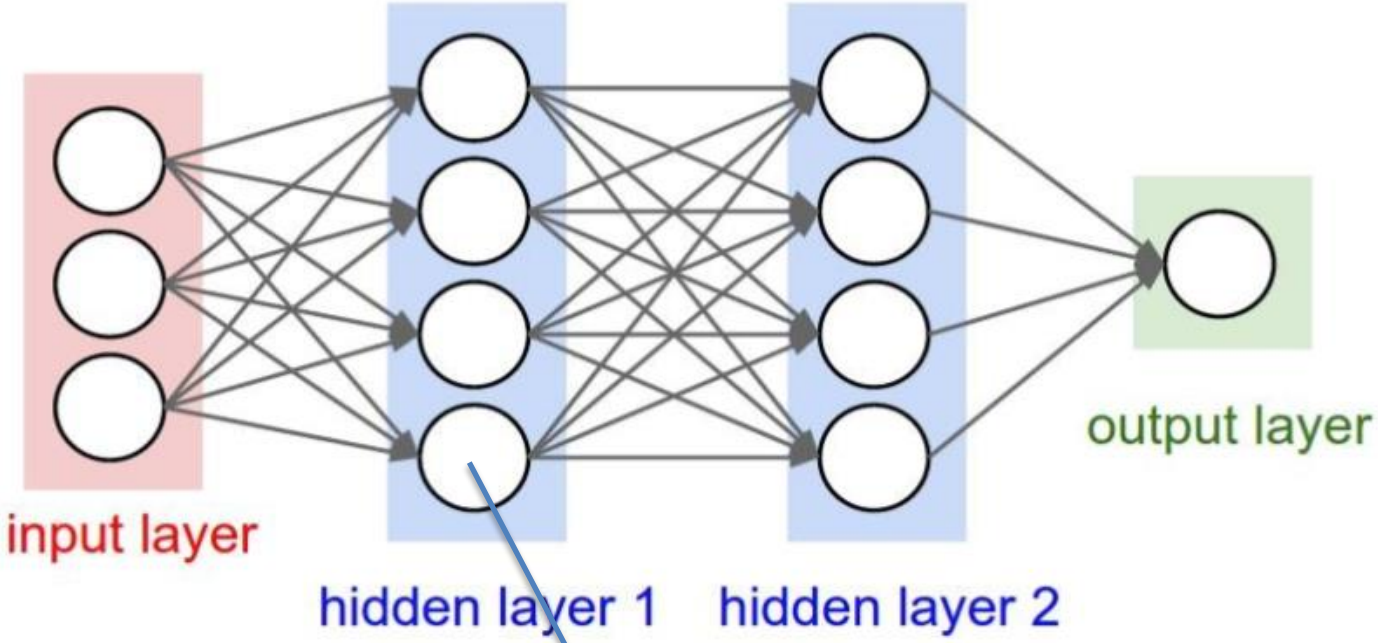
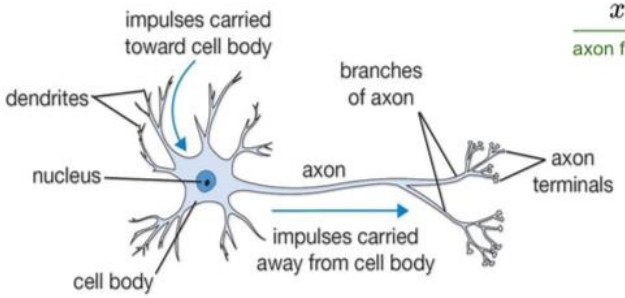
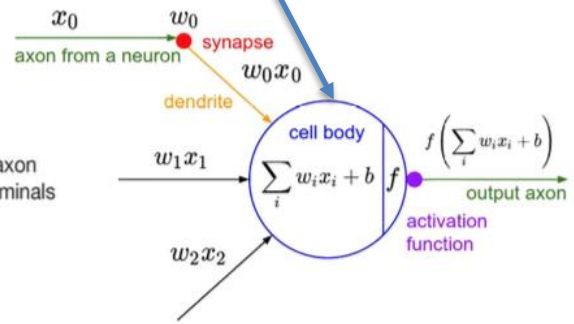
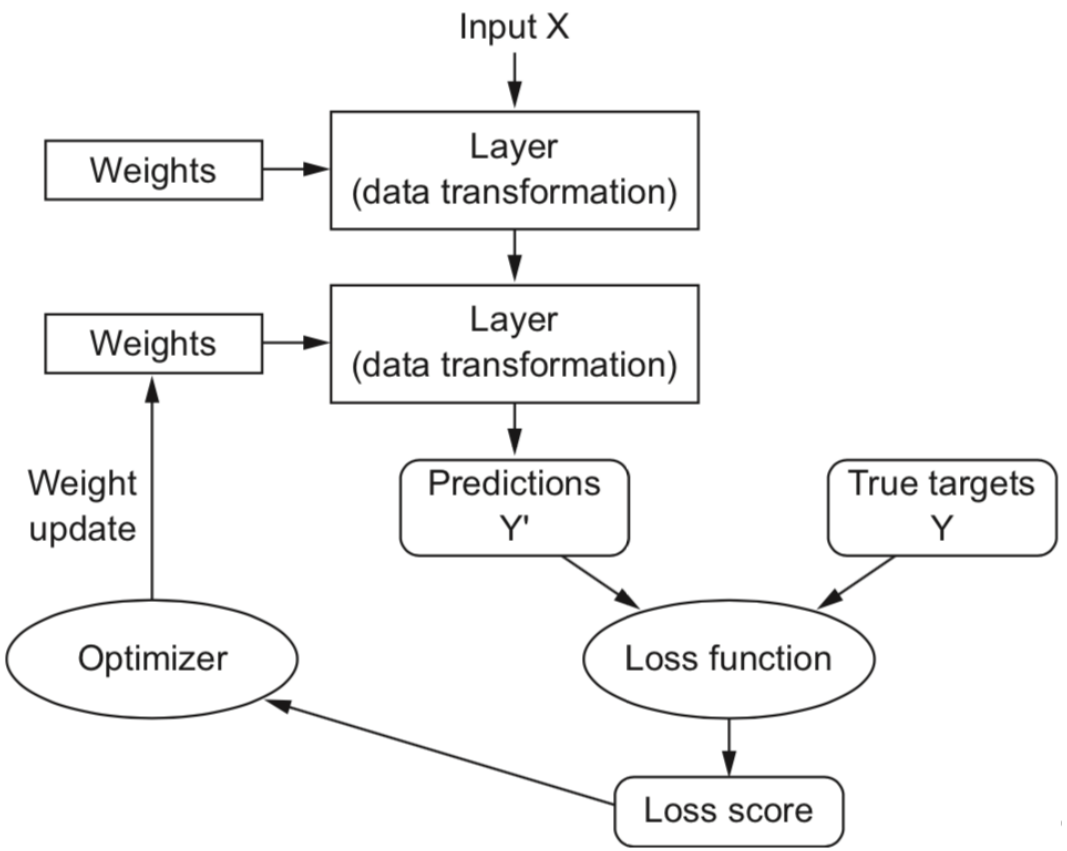
Wat is een neuraal netwerk?





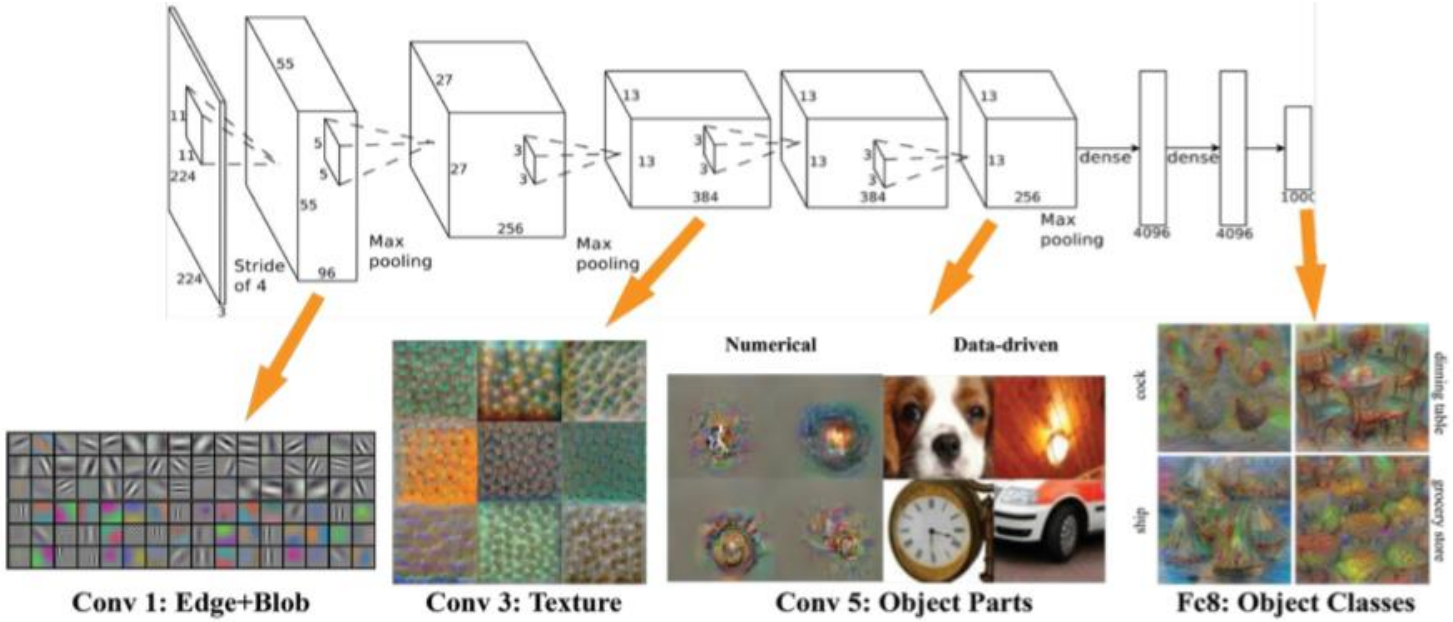


Hoe werkt een neuraal network?



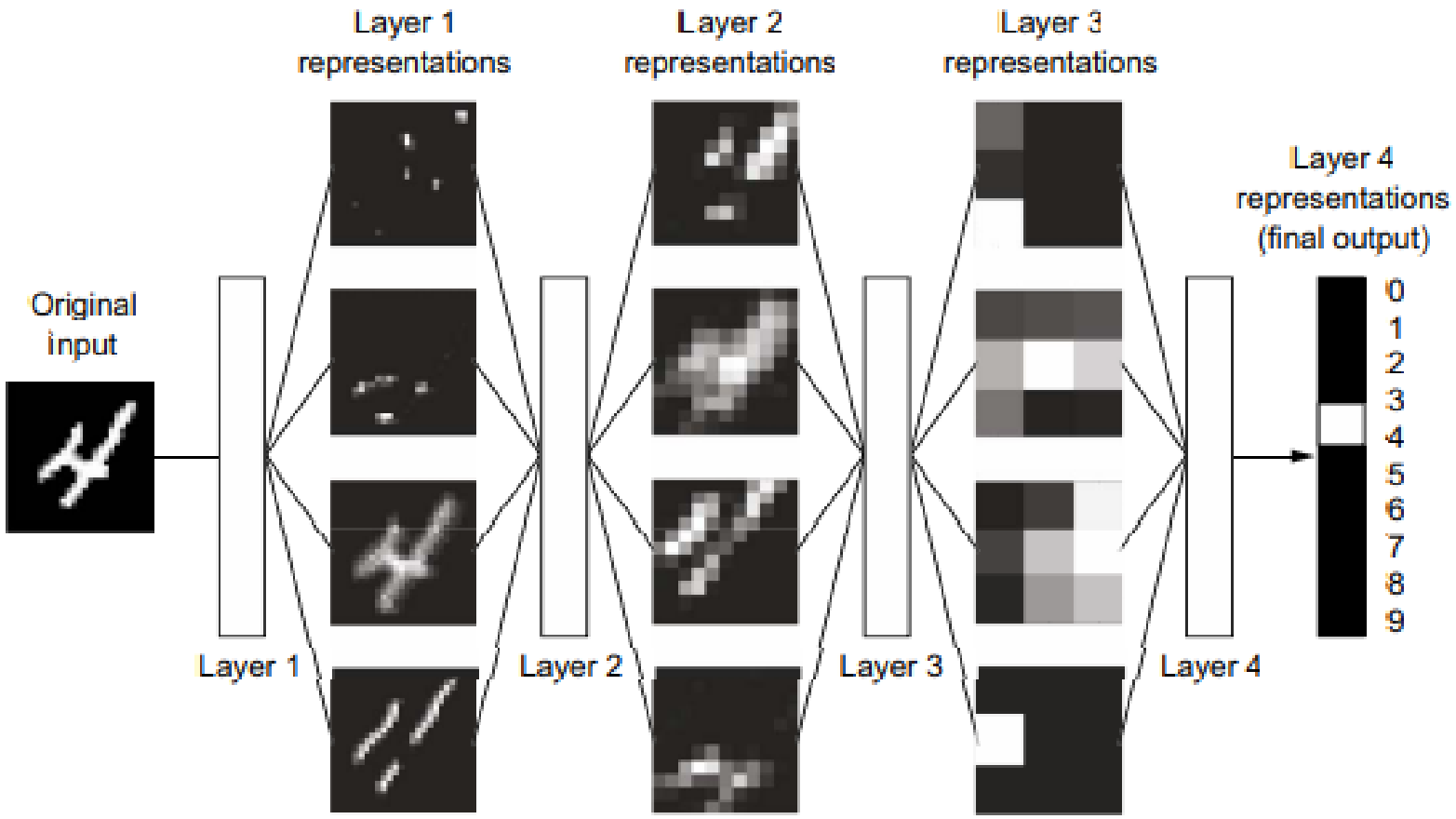
Convolutional Neural Networks

- Mainly **“visual” problems**, e.g. image classification
- A **Convolution layer** is a **feature detector** that automagically learns to filter out not needed information from an input by using a convolution kernel.
- Pooling layers compute the max or average value of a particular feature over a region of the input data (downsizing of input images). Also helps to detect objects in some unusual places and reduces memory size.



Voorbeeld: image classification

MNIST dataset: duizenden handgeschreven cijfers



Python

Waarom gebruiken we Python?

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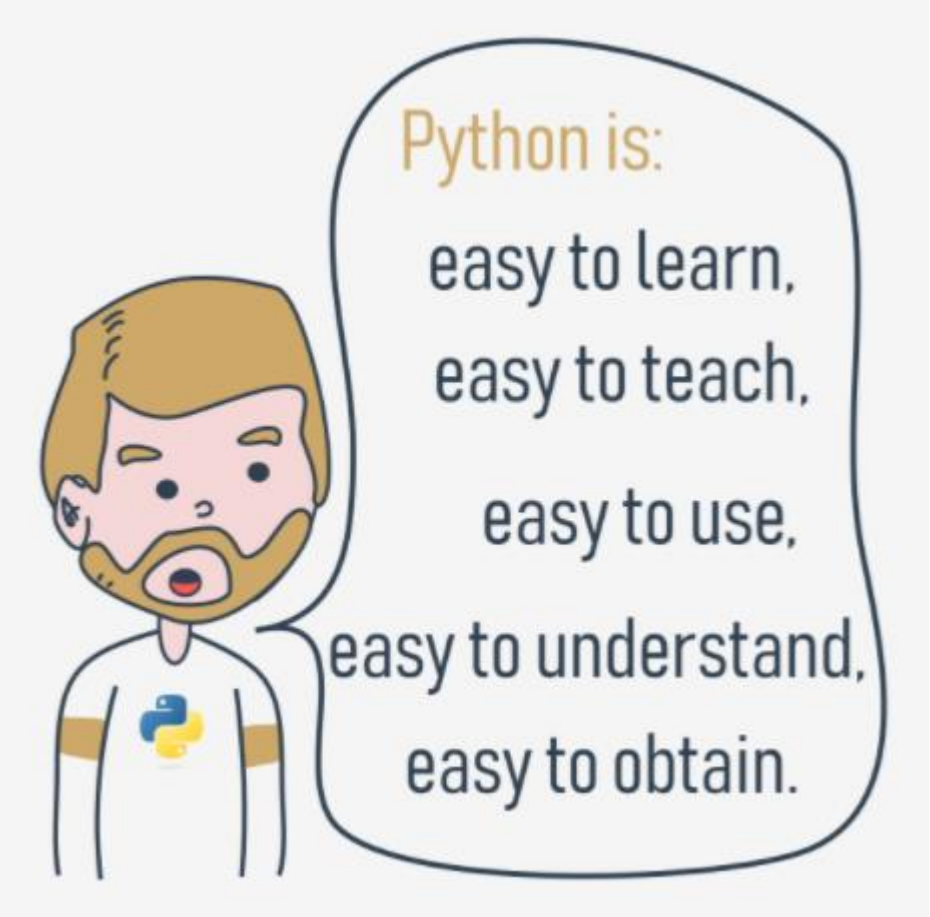
Waarom Python?



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R versus Python

Difference Between R and Python

Features	R	Python
Scope	Used mainly for statistical modeling	Used for a variety of purposes like web-application development and data analysis
Used By	Statisticians, Analyst & Data Scientist	Developer, Data Engineers & Data Scientist
Suitable For	People with no prior experience in programming	Newbies to experienced IT professionals
Package Distribution	CRAN	PyPi
Visualization Tools	ggplot2, plotly, ggiraph	Matplotlib, bokkeh, seaborn

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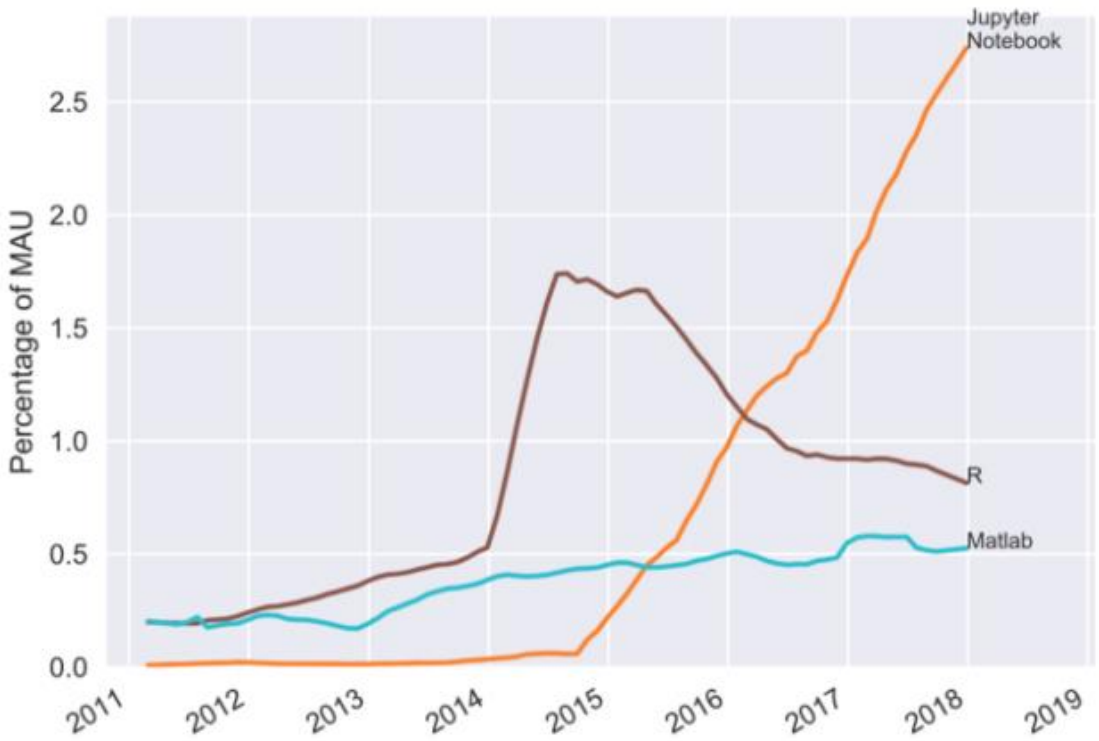
Jupyter notebooks vs R vs Matlab



"Ranking Programming Languages by GitHub Users" — Ben Frederickson

MAU = monthly active users

Populaire Python packages

seaborn

plotly

NumPy

theano

mxnet

jupyter

Keras

bokeh

NLTK

pandas

matplotlib

scikit learn

SciPy

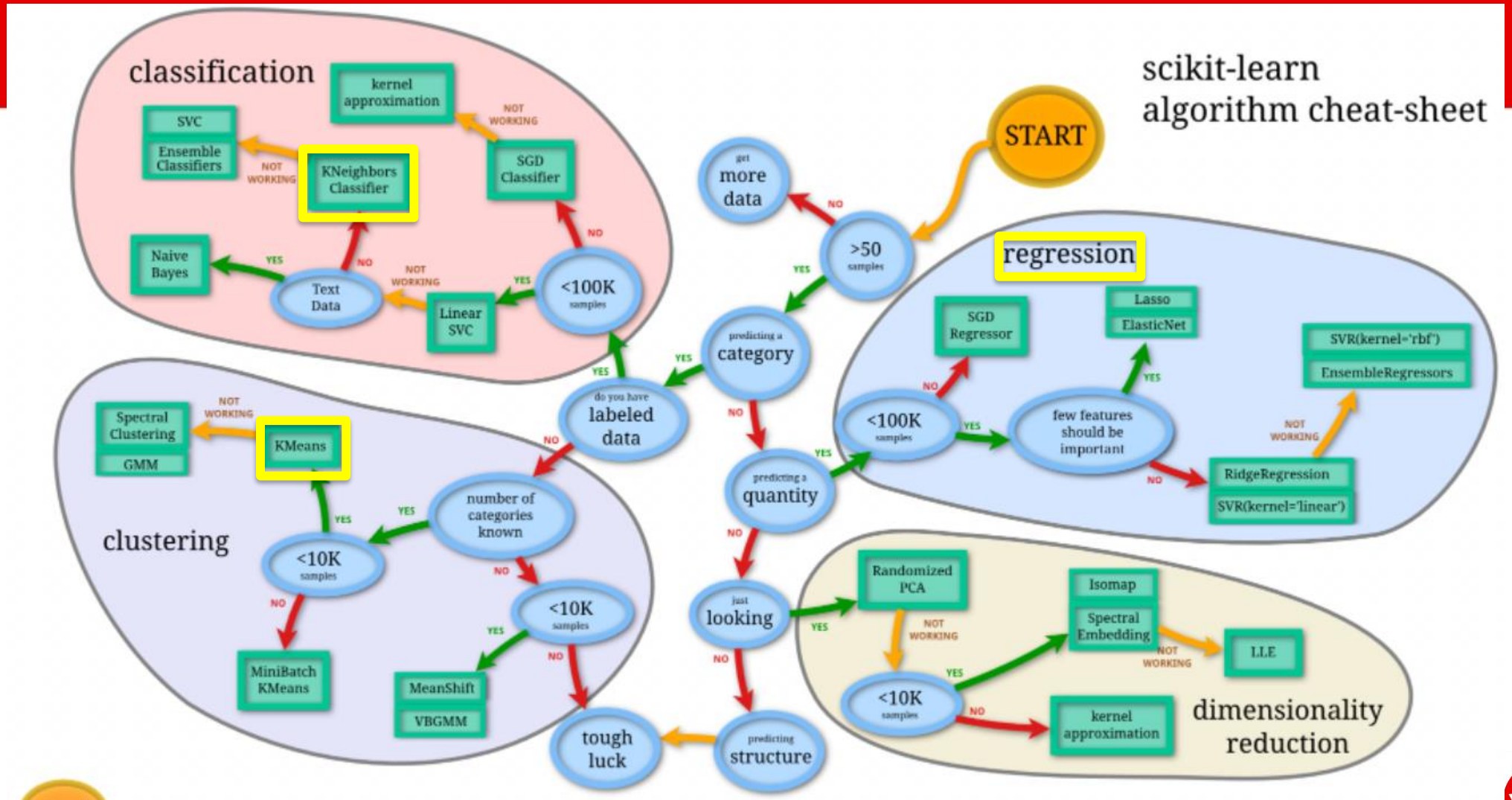
pillow

PYTORCH

APACHE Spark MLlib

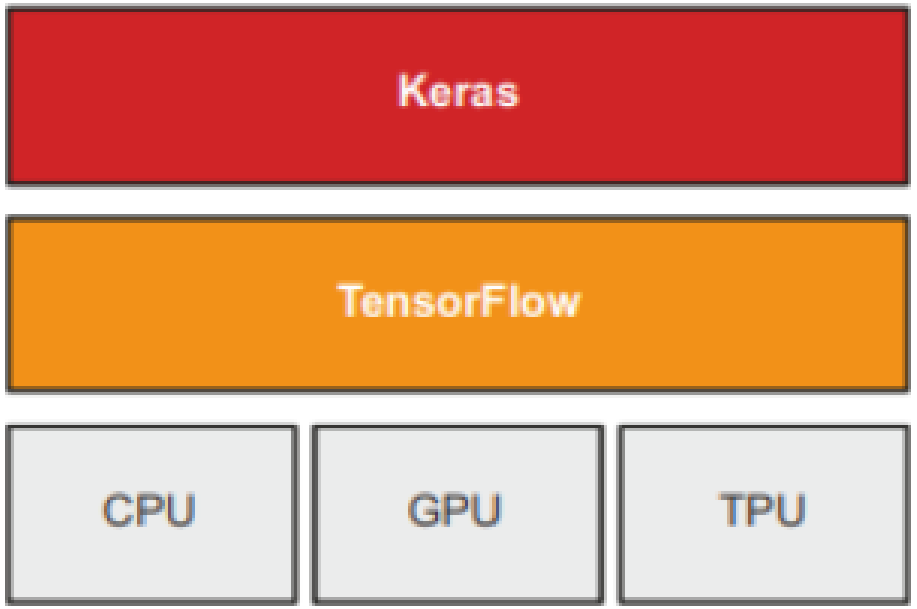
TensorFlow

Scikit-Learn



Back

TensorFlow en Keras



Deep learning development: layers, models, optimizers, losses, metrics...

Tensor manipulation infrastructure: tensors, variables, automatic differentiation, distribution...

Hardware: execution

Bronnen

- Leermateriaal van het vak "Machine Learning Fundamentals", gedoceerd door Andy Louwyck en Stefaan Haspeslagh tijdens het academiejaar 2020-2021 in het tweede jaar van de specialisatie Artificial Intelligence binnen de opleiding professionele Bachelor Toegepaste Informatica aan de Vives hogeschool van Kortrijk.
- Leermateriaal van het vak "Deep Learning Fundamentals and Advanced", gedoceerd door Andy Louwyck en Stefaan Haspeslagh tijdens het academiejaar 2021-2022 in het derde jaar van de specialisatie Artificial Intelligence binnen de opleiding professionele Bachelor Toegepaste Informatica aan de Vives hogeschool van Kortrijk.
- Serrano, L. G., 2020. Grokking Machine Learning (Version 12), Manning Early Access Program, Manning Publications, 500p.
- Chollet, F., 2021. Deep Learning with Python, 2nd edition, Manning Publications, 478p.

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